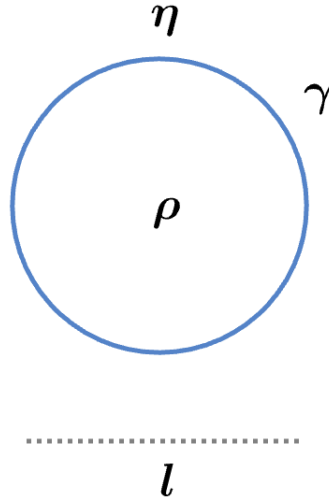


2024 F=ma Exam: Problem 15

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We can do dimensional analysis to find the Ohnesorge number $O = O(\eta, \rho, \gamma, l)$. We are given

$$[O] = 1$$

and $F = \eta Av/d$ so

$$[\eta] = \left[\frac{Fd}{Av} \right] = \frac{(ML/T^2)L}{L^2(L/T)} = \frac{ML^2}{T^2} \frac{T}{L^3} = \frac{M}{LT}$$

Also recall

$$\begin{aligned} [\rho] &= \frac{M}{L^3} \\ [\gamma] &= \frac{[F]}{[L]} = \frac{M}{T^2} \\ [l] &= L \end{aligned}$$

We are told $O \propto \eta$ so we need to divide by $\sqrt{\gamma}$ to cancel out dimensions of time,

$$\left[\frac{\eta}{\sqrt{\gamma}} \right] = \frac{M}{LT} \frac{T}{\sqrt{M}} = \frac{\sqrt{M}}{L}$$

Then we can divide by $\sqrt{\rho}$ to cancel out dimensions of mass,

$$\left[\frac{\eta}{\sqrt{\gamma \rho}} \right] = \frac{\sqrt{M}}{L} \frac{L^{3/2}}{\sqrt{M}} = \sqrt{L}$$

Finally we can divide by $\sqrt{l}$ to cancel out dimensions of length,

$$\left[\frac{\eta}{\sqrt{\gamma \rho l}} \right] = \frac{\sqrt{L}}{\sqrt{L}} = 1$$

Thus,

$$O \propto \frac{\eta}{\sqrt{\gamma \rho l}}$$

so the answer is E.